

Module 2 Lab Exercise: Tools Used in Machine Learning

Learning Objectives

By the end of this lab, you will be able to:

- Set up and navigate Jupyter Notebook, Google Colab, and VS Code environments
- Install and import essential Python libraries for machine learning
- Create and format professional documentation using Markdown
- Initialize a GitHub repository for your ML projects
- Understand the basic workflow of data science tools

Prerequisites

- Basic understanding of what machine learning is (Module 1)
- Access to internet for downloading tools and datasets
- A Google account (for Colab) or local Python installation

Part 1: Environment Setup and Tool Overview

What are the main tools we'll use in this course?

Jupyter Notebook/Google Colab: Interactive computing environments where you can write code, see results immediately, and document your work with text and visualizations.

Python Libraries: Pre-written code packages that make machine learning tasks easier:

- **Pandas:** For working with data (like Excel, but more powerful)
- **NumPy:** For mathematical operations on arrays of numbers
- **Matplotlib:** For creating charts and graphs
- **Scikit-learn:** The main library for machine learning algorithms

GitHub: A platform to store, share, and collaborate on code projects

VS Code: A powerful text editor for writing and debugging code

Let's start by setting up our environment!

✓ Environment Setup Instructions

Option 1: Google Colab (Recommended for Beginners)

1. Go to colab.research.google.com
2. Sign in with your Google account
3. Click "New Notebook"
4. You're ready to go! Libraries are pre-installed.

Option 2: Local Jupyter Notebook

1. Install Python from python.org
2. Open terminal/command prompt
3. Run: `pip install jupyter pandas numpy matplotlib scikit-learn`
4. Run: `jupyter notebook`
5. Create a new notebook

Option 3: VS Code

1. Download VS Code from code.visualstudio.com
2. Install Python extension
3. Install Jupyter extension
4. Create a new .ipynb file

For this lab, we recommend starting with Google Colab as it requires no installation.

```
# Install required libraries (uncomment if needed)
# !pip install pandas numpy matplotlib scikit-learn

# Import libraries with standard aliases
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn import datasets
import warnings
warnings.filterwarnings('ignore') # Hide warning messages for cleaner output

print("✅ All libraries imported successfully!")
print(f"Pandas version: {pd.__version__}")
print(f"NumPy version: {np.__version__}")
```

```
✅ All libraries imported successfully!
Pandas version: 2.2.3
NumPy version: 2.1.3
```

✓ Part 2: Loading and Exploring Your First Dataset

We'll use the famous Iris dataset - a classic dataset for beginners. It contains measurements of iris flowers from three different species.

```
# Load a simple dataset (Iris flowers - a classic beginner dataset)
from sklearn.datasets import load_iris
```

```
# Load the data
iris = load_iris()
print("Dataset loaded successfully!")
print(f"Dataset shape: {iris.data.shape}")
print(f"Features: {iris.feature_names}")
print(f"Target classes: {iris.target_names}")
```

```
Dataset loaded successfully!
Dataset shape: (150, 4)
Features: ['sepal length (cm)', 'sepal width (cm)', 'petal length (cm)', 'petal
Target classes: ['setosa' 'versicolor' 'virginica']
```

```
# Convert to pandas DataFrame for easier handling
df = pd.DataFrame(iris.data, columns=iris.feature_names)
df['species'] = iris.target_names[iris.target]
```

```
# Display first few rows
print("First 5 rows of our dataset:")
print(df.head())
```

```
print("\nDataset info:")
print(df.info())
```

```
First 5 rows of our dataset:
   sepal length (cm)  sepal width (cm)  petal length (cm)  petal width (cm)  \
0                5.1                3.5                1.4                0.2
1                4.9                3.0                1.4                0.2
2                4.7                3.2                1.3                0.2
3                4.6                3.1                1.5                0.2
4                5.0                3.6                1.4                0.2
```

```
   species
0  setosa
1  setosa
2  setosa
3  setosa
4  setosa
```

```
Dataset info:
<class 'pandas.core.frame.DataFrame'>
```

```

RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column                Non-Null Count  Dtype  
---  -
 0   sepal length (cm)      150 non-null    float64
 1   sepal width (cm)       150 non-null    float64
 2   petal length (cm)      150 non-null    float64
 3   petal width (cm)       150 non-null    float64
 4   species                150 non-null    object  
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
None

```

✓ Part 3: Creating Your First Visualization

Data visualization is crucial in machine learning. Let's create a simple plot to understand our data.

```

# Create a simple scatter plot
plt.figure(figsize=(10, 6))

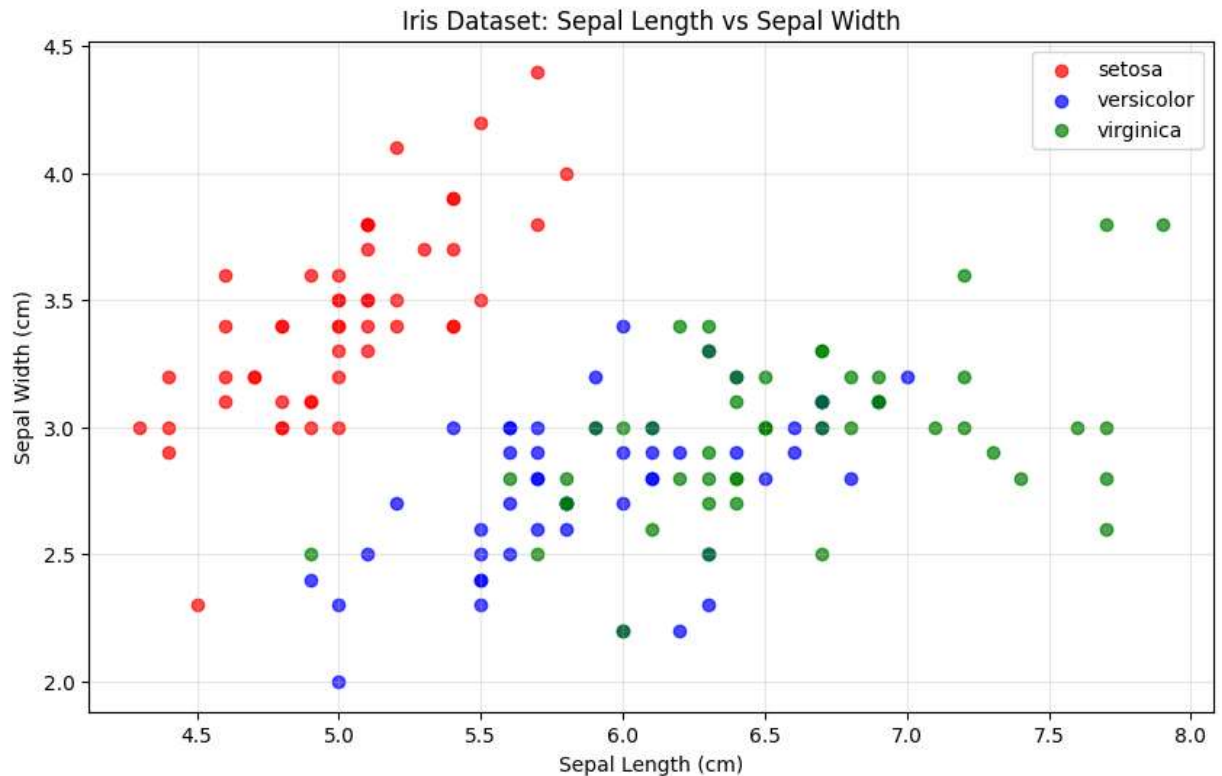
# Plot sepal length vs sepal width, colored by species
species_colors = {'setosa': 'red', 'versicolor': 'blue', 'virginica': 'green'}

for species in df['species'].unique():
    species_data = df[df['species'] == species]
    plt.scatter(species_data['sepal length (cm)'],
                species_data['sepal width (cm)'],
                c=species_colors[species],
                label=species,
                alpha=0.7)

plt.xlabel('Sepal Length (cm)')
plt.ylabel('Sepal Width (cm)')
plt.title('Iris Dataset: Sepal Length vs Sepal Width')
plt.legend()
plt.grid(True, alpha=0.3)
plt.show()

print("🎉 Congratulations! You've created your first data visualization!")

```



🎉 Congratulations! You've created your first data visualization!

✓ Part 4: Practice with Basic Data Operations

Let's practice some basic data analysis operations that you'll use throughout the course.

```
# Basic statistical analysis
print("Basic Statistics for Iris Dataset:")
print("=" * 40)

# Calculate mean values for each species
species_means = df.groupby('species').mean()
print("\nMean values by species:")
print(species_means)

# Count samples per species
species_counts = df['species'].value_counts()
print("\nSamples per species:")
print(species_counts)
```

Basic Statistics for Iris Dataset:

=====

Mean values by species:

	sepal length (cm)	sepal width (cm)	petal length (cm)	\
species				
setosa	5.006	3.428	1.462	
versicolor	5.936	2.770	4.260	
virginica	6.588	2.974	5.552	

	petal width (cm)
species	
setosa	0.246
versicolor	1.326
virginica	2.026

Samples per species:

species	
setosa	50
versicolor	50
virginica	50

Name: count, dtype: int64

Part 5: GitHub and Documentation Best Practices

Why GitHub for Machine Learning?

- **Version Control:** Track changes to your code and data
- **Collaboration:** Work with others on projects
- **Portfolio:** Showcase your work to potential employers
- **Backup:** Never lose your work

Basic GitHub Workflow:

1. **Create Repository:** A folder for your project
2. **Clone/Download:** Get the project on your computer
3. **Add Files:** Put your notebooks and data
4. **Commit:** Save a snapshot of your changes
5. **Push:** Upload changes to GitHub

For This Course:

- Create a repository named "ITAI-1371-ML-Labs"
- Upload each lab notebook as you complete it
- Include a README.md file describing your projects

Action Item: After this lab, create your GitHub account and repository.

✓ Assessment: Tool Familiarity Check

Complete the following tasks to demonstrate your understanding of the tools:

```
# Task 1: Create a simple calculation using NumPy
# Calculate the mean and standard deviation of sepal length

sepal_lengths = df['sepal length (cm)']

# Your code here:
mean_sepal_length = np.mean(sepal_lengths)
std_sepal_length = np.std(sepal_lengths)

print(f"Mean sepal length: {mean_sepal_length:.2f} cm")
print(f"Standard deviation: {std_sepal_length:.2f} cm")

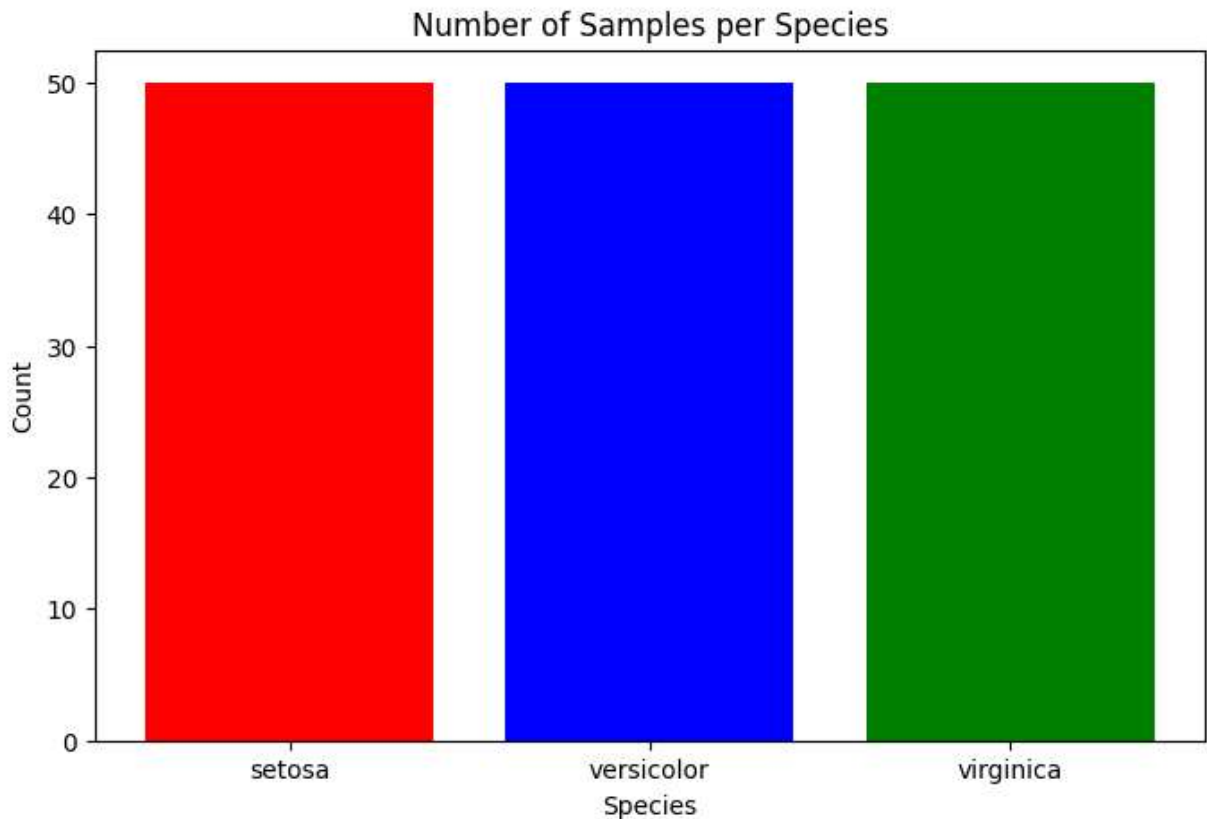
# Verification (don't modify)
assert isinstance(mean_sepal_length, (float, np.floating)), "Mean should be a r
assert isinstance(std_sepal_length, (float, np.floating)), "Std should be a nur
print("✓ Task 1 completed successfully!")
```

```
Mean sepal length: 5.84 cm
Standard deviation: 0.83 cm
✓ Task 1 completed successfully!
```

```
# Task 2: Create a simple bar chart showing species counts
species_counts = df['species'].value_counts()

plt.figure(figsize=(8, 5))
plt.bar(species_counts.index, species_counts.values, color=['red', 'blue', 'gre
plt.title('Number of Samples per Species')
plt.xlabel('Species')
plt.ylabel('Count')
plt.show()

print(f"Species distribution: {dict(species_counts)}")
print("✓ Task 2 completed successfully!")
```



Species distribution: {'setosa': np.int64(50), 'versicolor': np.int64(50), 'virginica': np.int64(50)}

✓ Task 2 completed successfully!

Your Analysis and Reflection

Instructions: Complete the analysis below by editing this markdown cell.

My Observations About the Iris Dataset

Dataset Overview:

- Number of samples: [150]
- Number of features: [4]
- Number of classes: [3]

Key Findings from the Visualization:

1. Setosa forms an isolated, cleanly separable cluster based on sepal dimensions.
2. Versicolor & Virginica exhibit overlapping distributions, sepals cannot reliably separate the two species..
3. Virginica flowers tend to have larger sepal lengths and widths compared to Versicolor.

Questions for Further Investigation:

- How much separation do petal length and width add across all three species??
- What classification accuracy does a simple model hit using only sepal features??
- Would dimensionality reduction or parallel coordinates reveal clearer decision boundaries?

Reflection: *Set up my ML environment in Colab, loaded a dataset with Pandas, and made a scatter plot in Matplotlib. Learned how to read basic stats and actually spot patterns in the data instead of just looking at numbers. Good baseline for future labs.*

Note: This is practice for documenting your machine learning projects professionally.

Lab Summary and Next Steps

What You've Accomplished:

- ✓ Set up your machine learning development environment
- ✓ Imported and used essential Python libraries
- ✓ Loaded and explored your first dataset
- ✓ Created your first data visualization
- ✓ Practiced professional documentation with Markdown
- ✓ Learned about GitHub for project management

Preparation for Module 3:

In the next lab, you'll:

- Learn about different types of machine learning
- Build your first simple classifier
- Understand the complete ML workflow
- Work with more complex datasets

Action Items:

1. **Create your GitHub account** and repository
2. **Upload this completed notebook** to your repository
3. **Experiment** with different visualizations using the Iris dataset
4. **Practice** Markdown formatting in a new notebook

Resources for Continued Learning:

- [Pandas Documentation](#)

- [Matplotlib Gallery](#)
- [GitHub Guides](#)
- [Jupyter Notebook Tips](#)